Data-driven reconstruction of large scales in atmospheric boundary layer flow from sparse SPIV/HWA measurements

International Workshop on Physical Modelling of Flow and Dispersion Phenomena – PHYSMOD 2026

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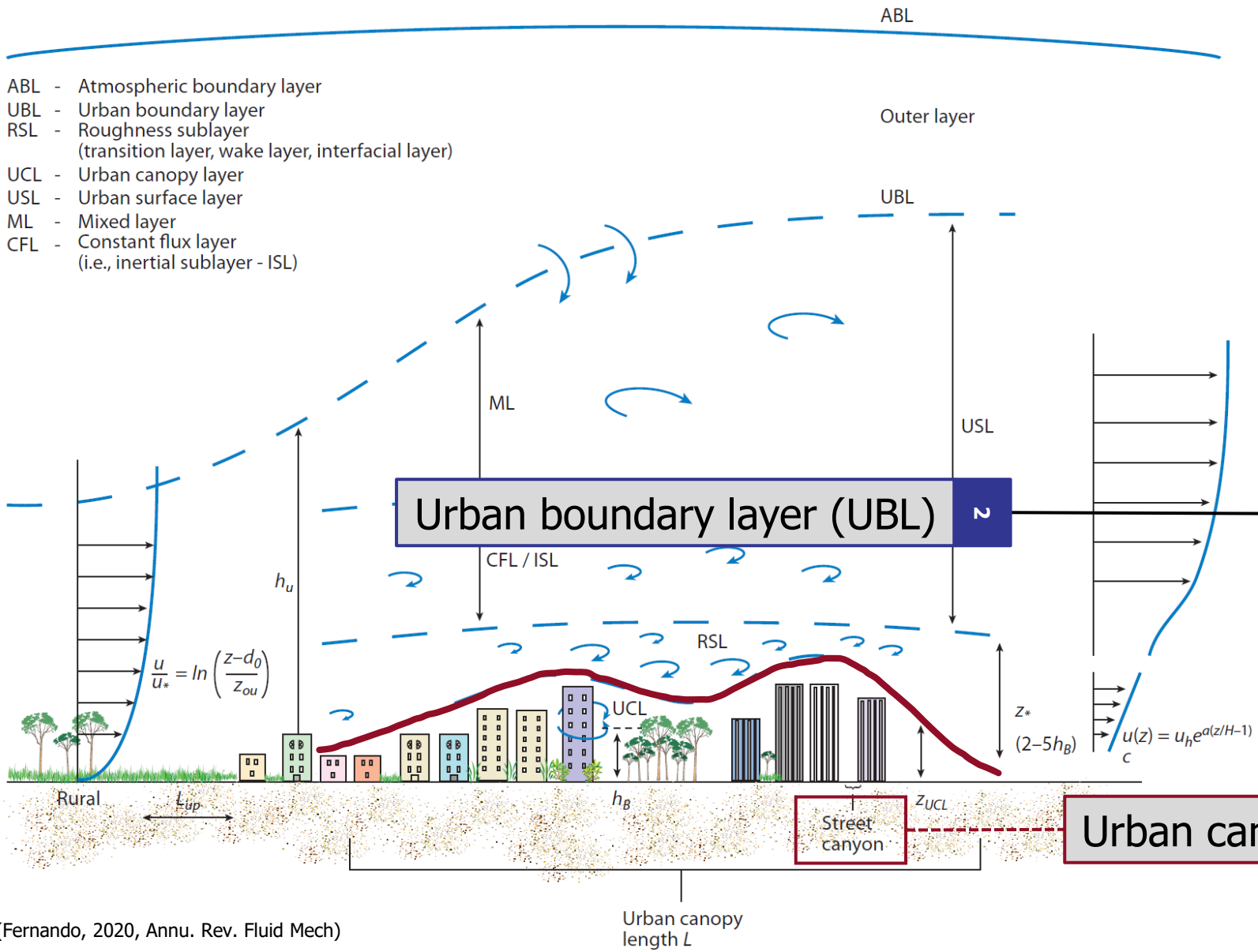
Haoran Du*, Eric Savory, Laurent Perret

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Background

> Background > General Information



(Fernando, 2020, Annu. Rev. Fluid Mech)

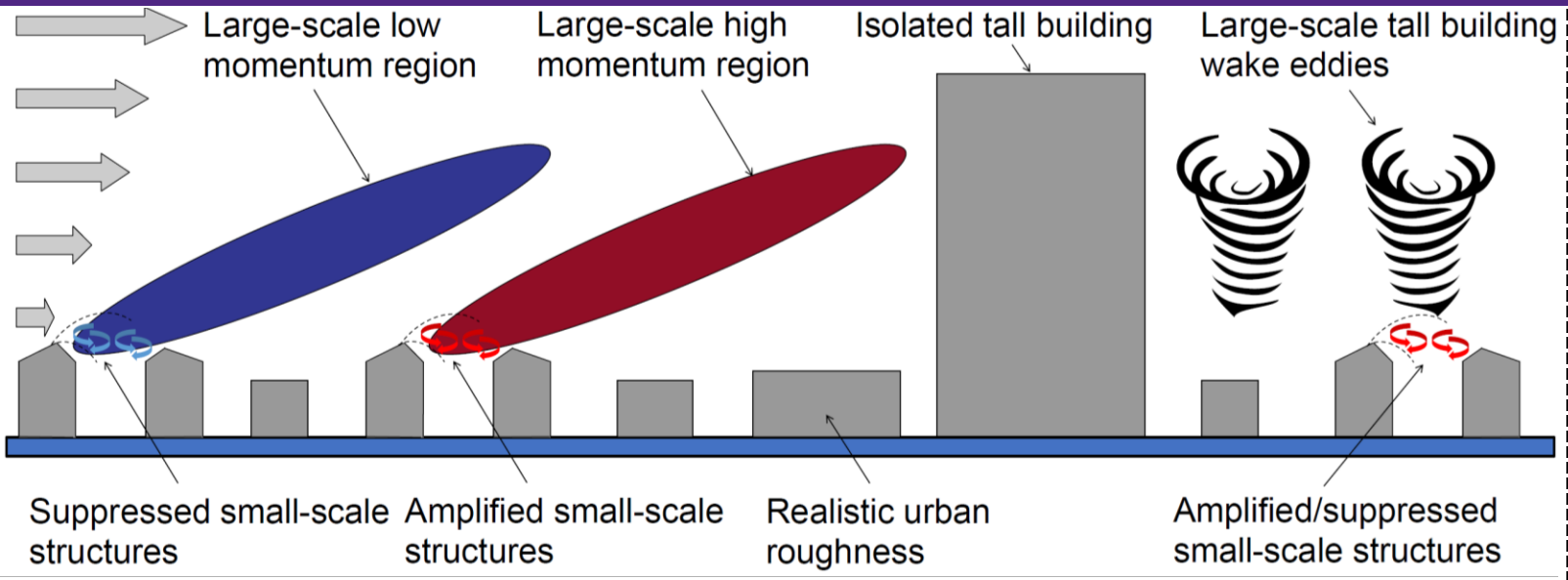
Urban boundary layer (UBL) 2

Urban canopy layer (UCL) 1

Dynamic transport of air and pollutants

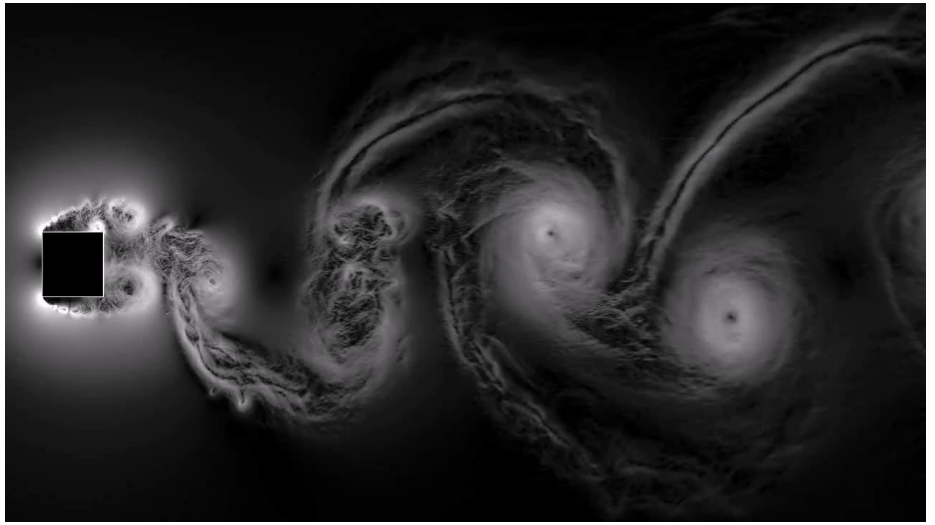


> Background > Coherent Structures in ABL



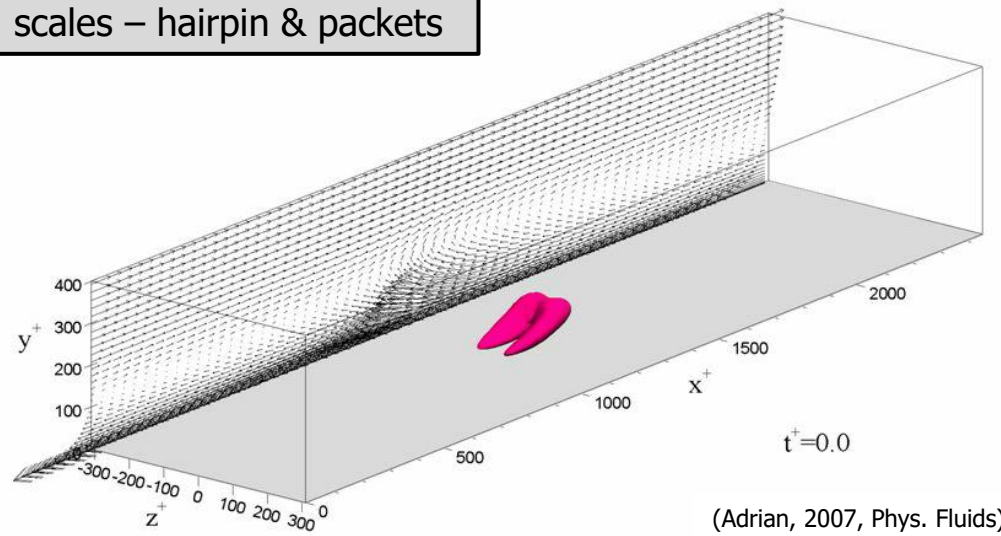
(Du et al., 2024, Bound.-Layer Meteorol.)

Von Kármán vortex street of tall square cylinder



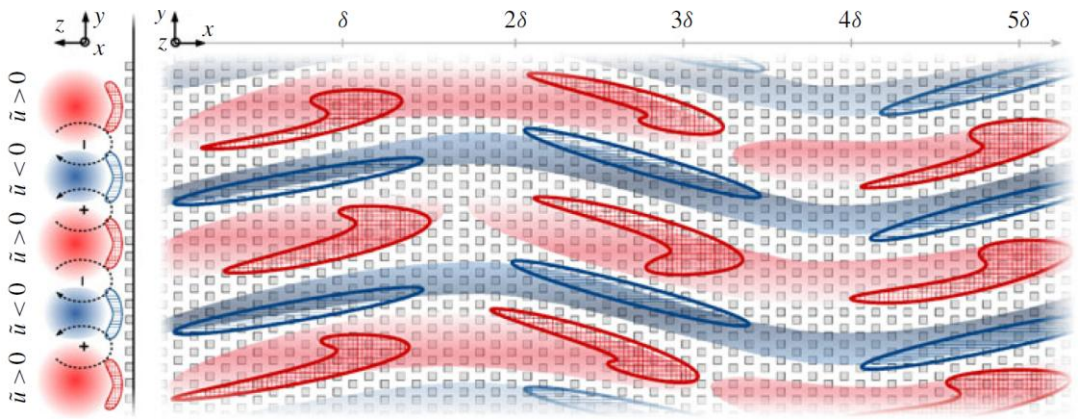
(Trias et al., 2015, Comput. Fluids)

Small scales – hairpin & packets



(Adrian, 2007, Phys. Fluids)

Very-large-scale motions (VLSMs)

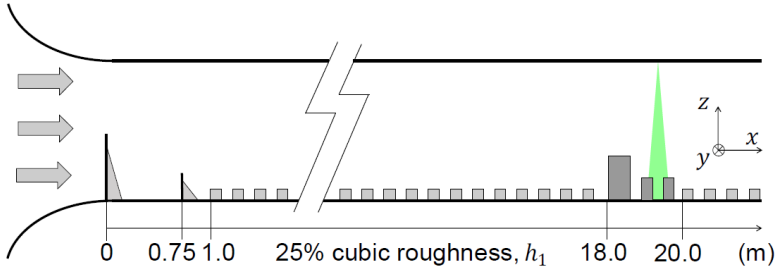


(Basley et al., 2018, J. Fluid Mech.)

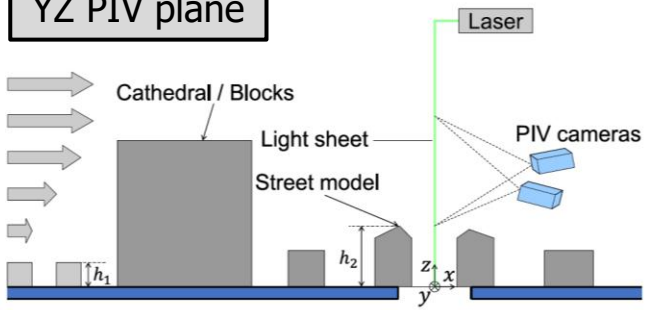
Datasets

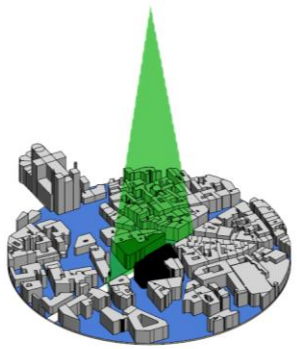
> Datasets > Wind Tunnel Experiment Campaigns

Morphological model Rue de Strasbourg (RdS) dataset

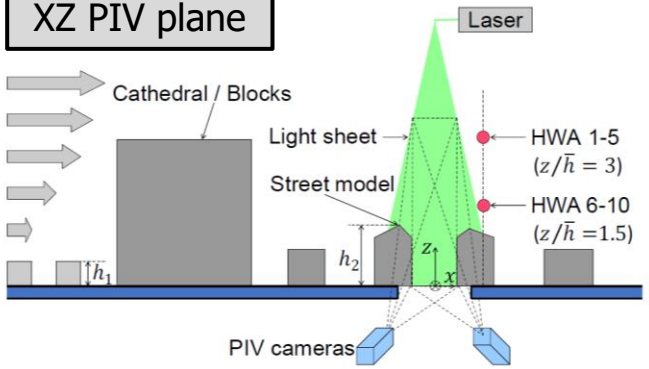


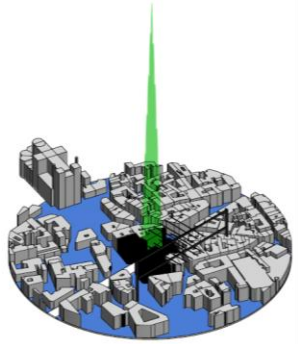
YZ PIV plane





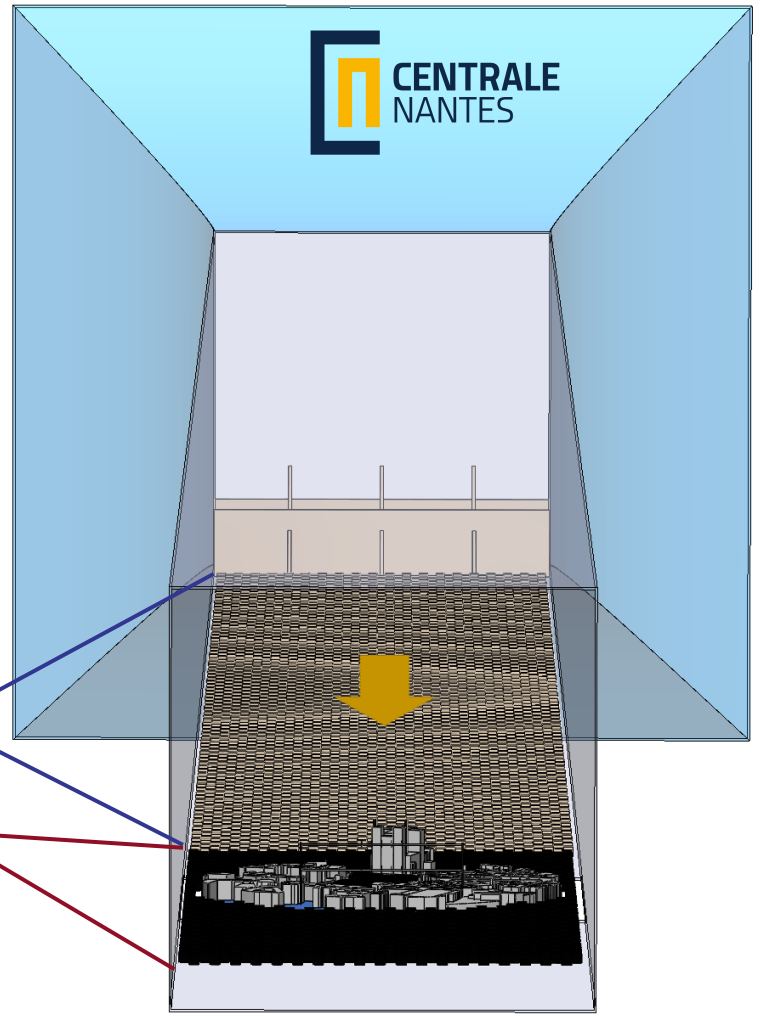
XZ PIV plane





- 1:200 scaled model
- 5 cases:
  - 1/2/3/6 stacked blocks
  - Cathedral
- Block size:
  - 0.2 m × 0.2 m × 0.1 m
- YZ & XZ planes (separately)

- Dimension before test section:
  - 18 m (x) × 2 m (y) × 2 m (z)
- Dimension of test section:
  - 2 m (x) × 2 m (y) × 2 m (z)
- Free-stream velocity: 5.8 m/s
- Friction velocity: 0.37 m/s

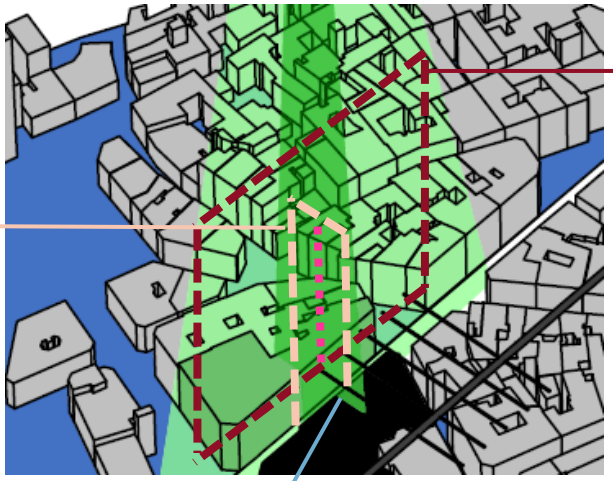


(Du et al., 2023, Build & Environ.)

> Datasets > Difficulty

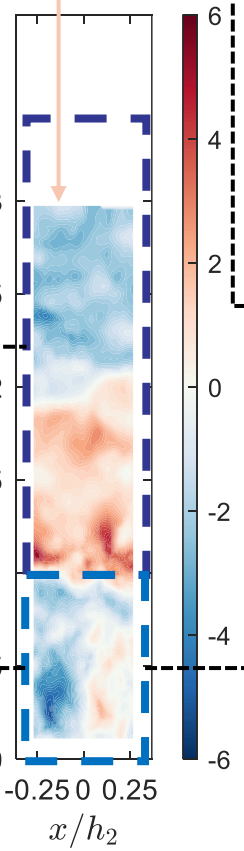
Areas in navy boxes can have correlations.

NOT synchronized



YZ PIV plane

XZ PIV plane

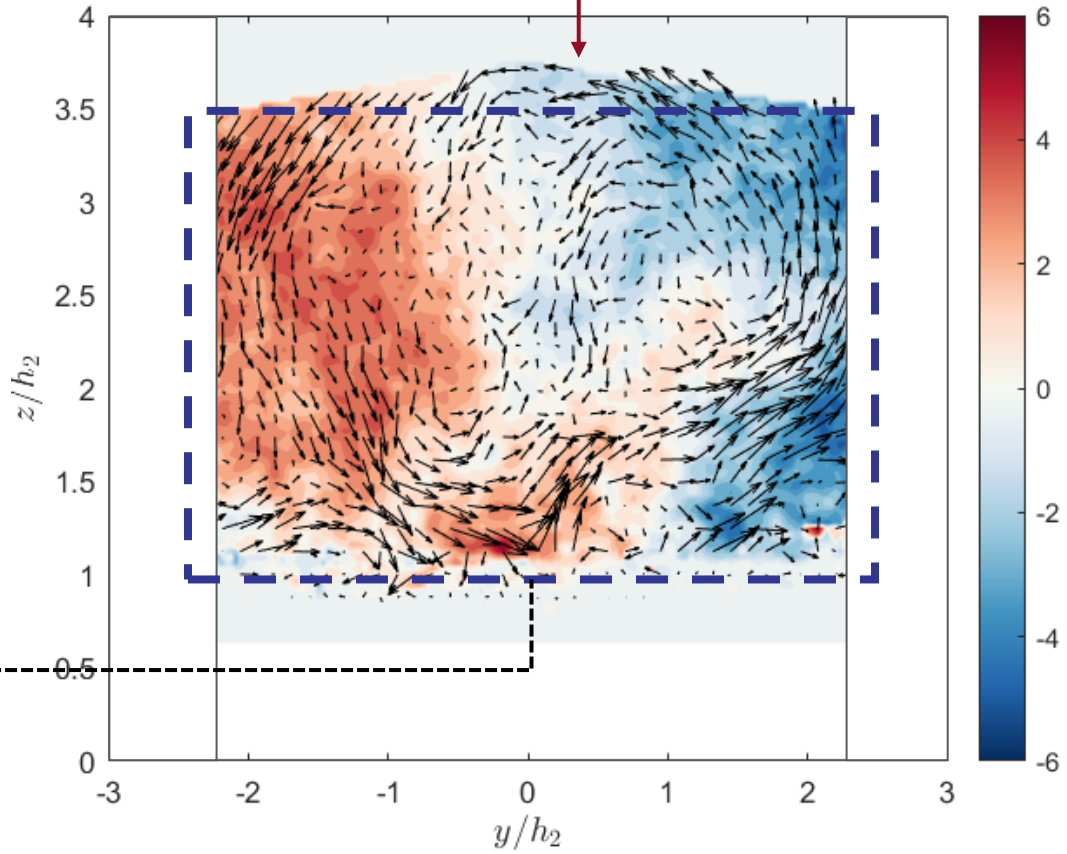


Synchronized

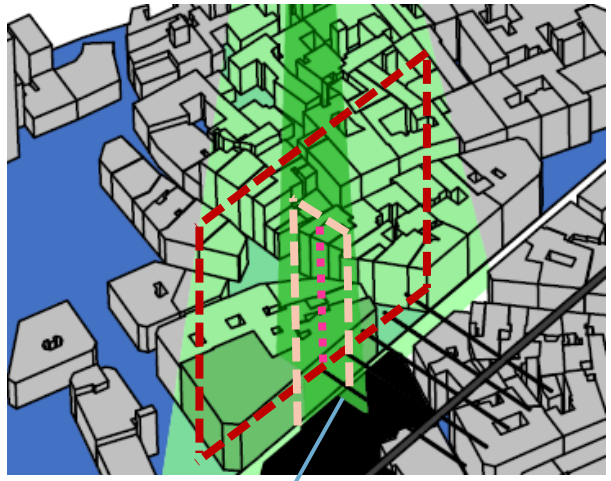
HWA probes (sensor-like)

Explored, Scale interaction

Correlations of XZ ($z/h_2 < 1$) and YZ ($z/h_2 > 1$)? Needs synchronization! Hard for experiments



> Datasets > Modelling idea



XZ PIV plane

YZ PIV plane

Synchronized

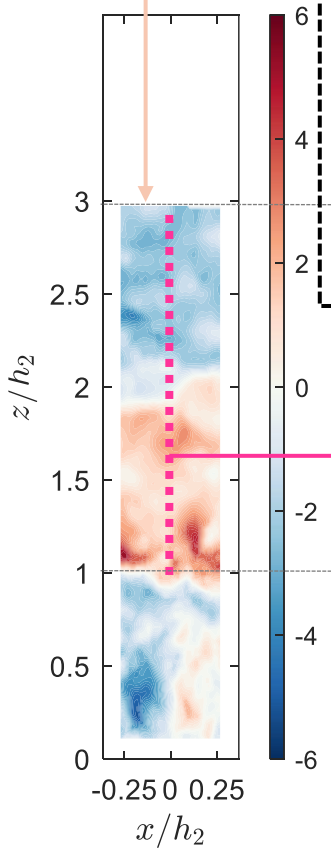
HWA probes (sensor-like)

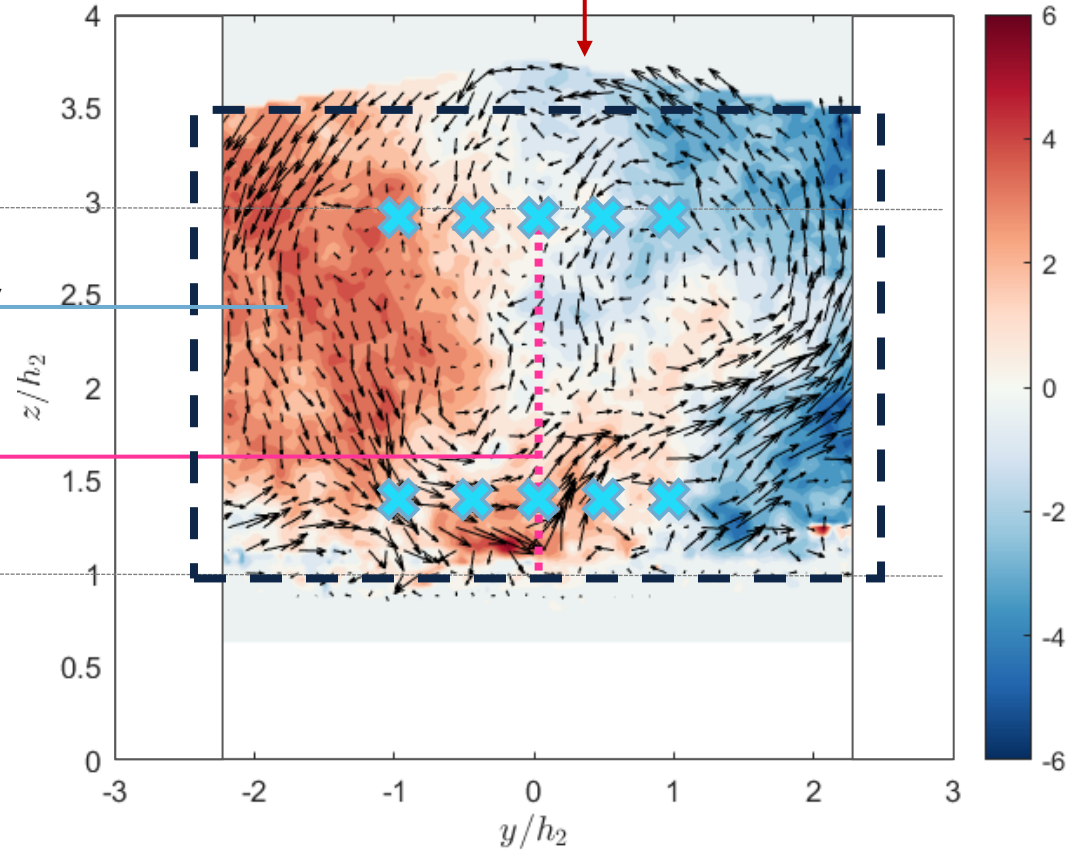
Extracted,

Synthetic

Intersection line

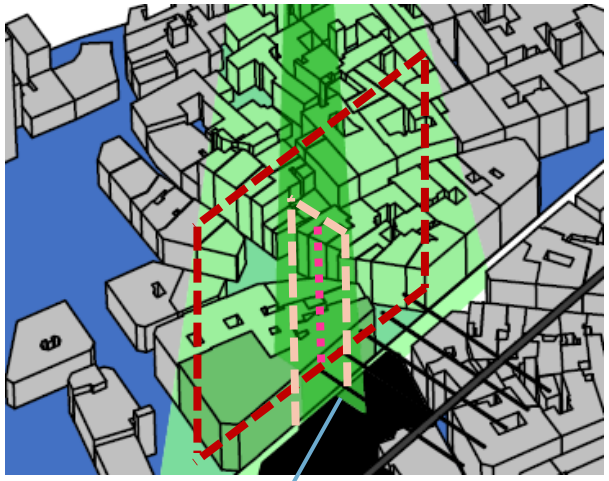
extracted from PIV planes





> Datasets > Input & Output & Research Question

1. Can we reconstruct the flow field from only sparse measurements?
2. Do more sophisticated nonlinear methods actually provide useful additional information?



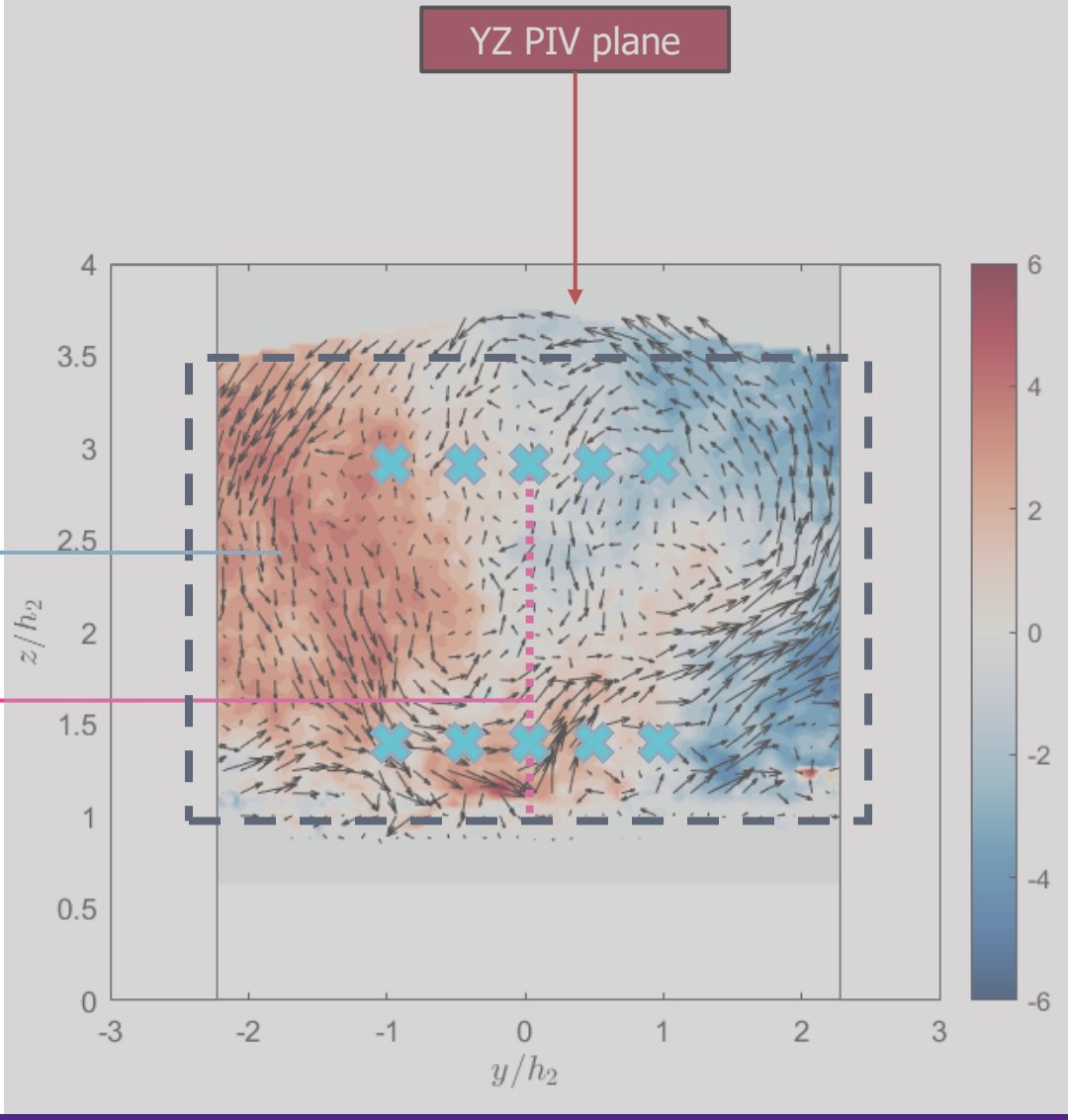
HWA probes (sensor-like)

Extracted,

Synthetic

Intersection line

extracted from PIV planes

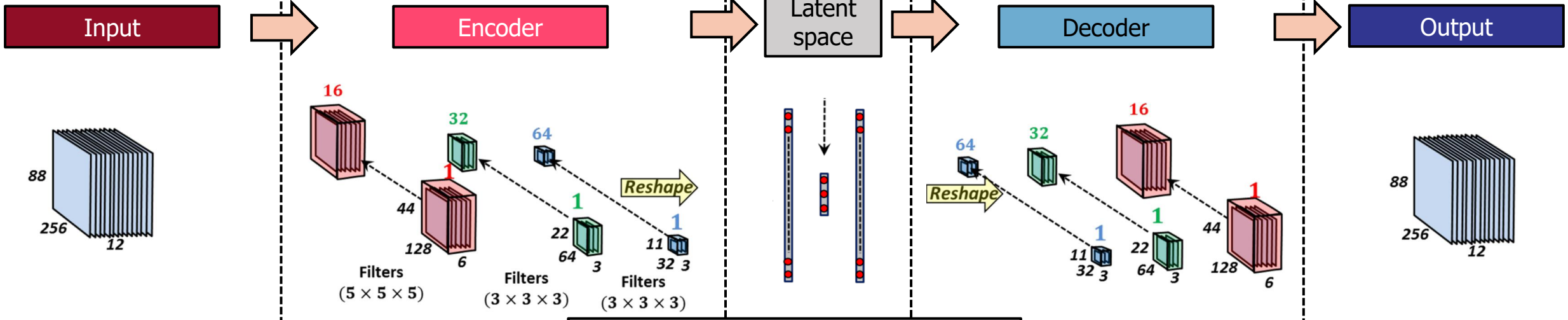


Methods

Why reduced-order latent space?

- Lower dimensionality.
- Better focus on dominant coherent structures.
- More efficient and stable reconstruction/prediction.

> Methods > Reduced-order representations



(Agostini, 2020, Phys. Fluids)

Convolutional Autoencoder (convAE)

x	$g_\theta(x)$	z	$h_\psi(z)$	$\hat{x}$

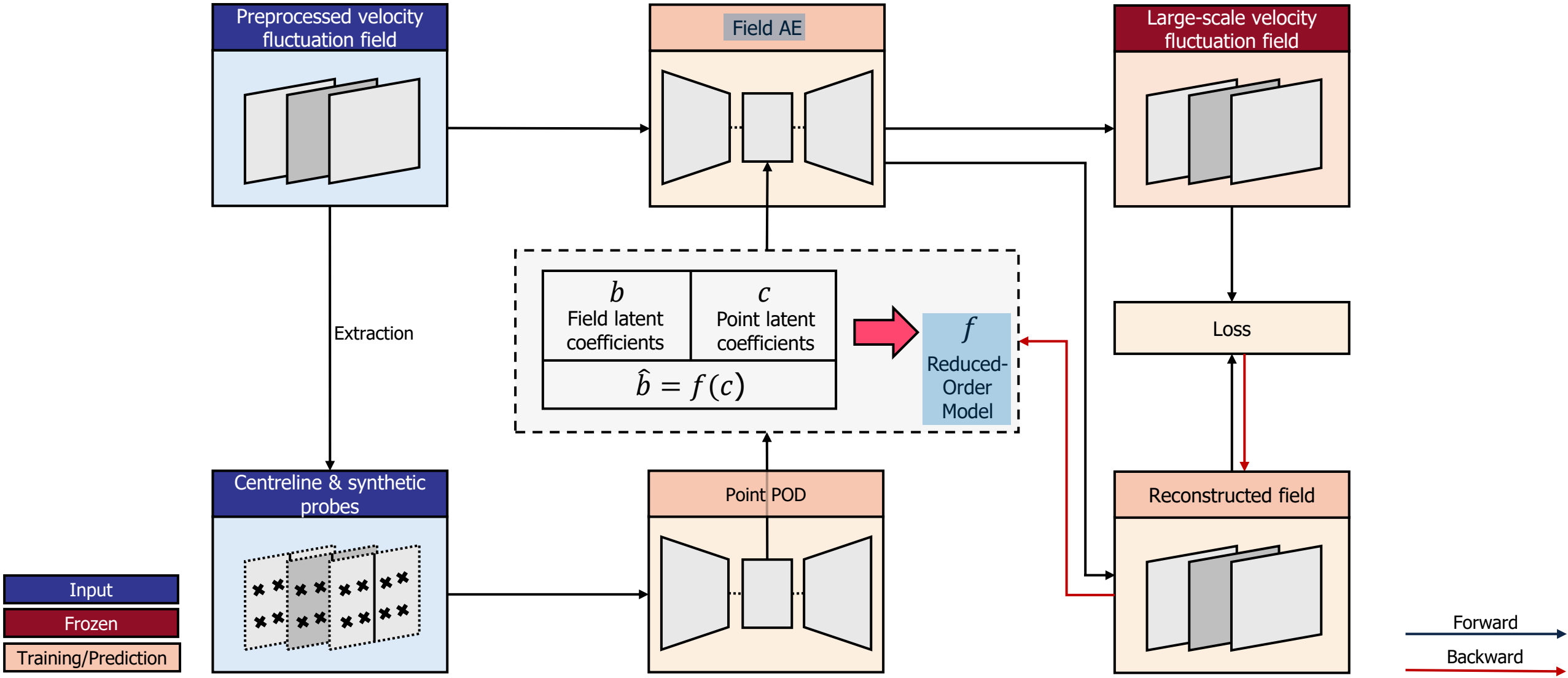
Proper Orthogonal Decomposition (POD)

Demean & Vectorise each snapshot field: $x \in \mathbb{R}^N$	$z = W_e x$, where $W_e = \mathbf{\Phi}_r^T \in \mathbb{R}^{r \times N}$ $\mathbf{\Phi}_r = [\Phi_1, \dots, \Phi_r]$ orthogonal modes	z	$\hat{x} = W_d z$, where $W_d = \mathbf{\Phi}_r \in \mathbb{R}^{N \times r}$	$\hat{x}$

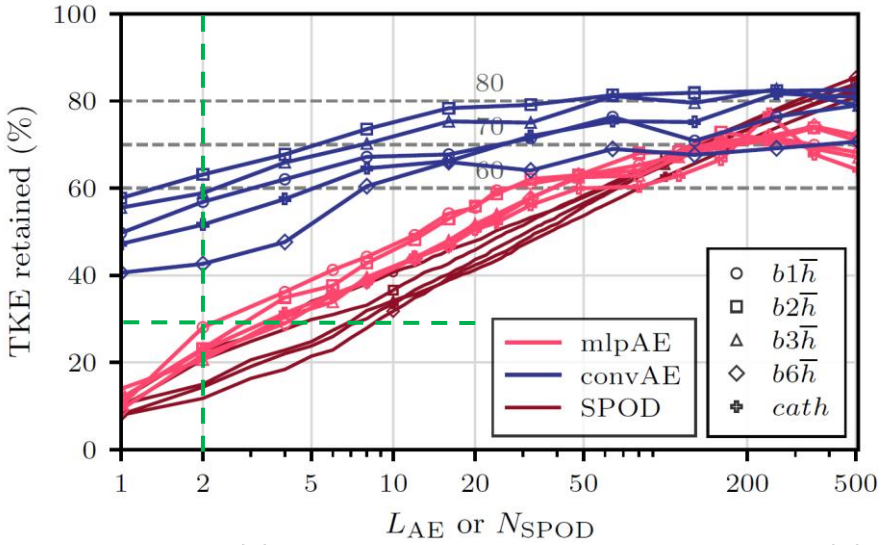
- POD/Spectral POD (SPOD): + interpretable, energy/frequency ranked, − linear only, may need many modes for nonlinear flow.
- AE: + nonlinear, compact latent for strong reconstruction; − less interpretable, not ranked, needs physics validation.

> Methods > Workflow

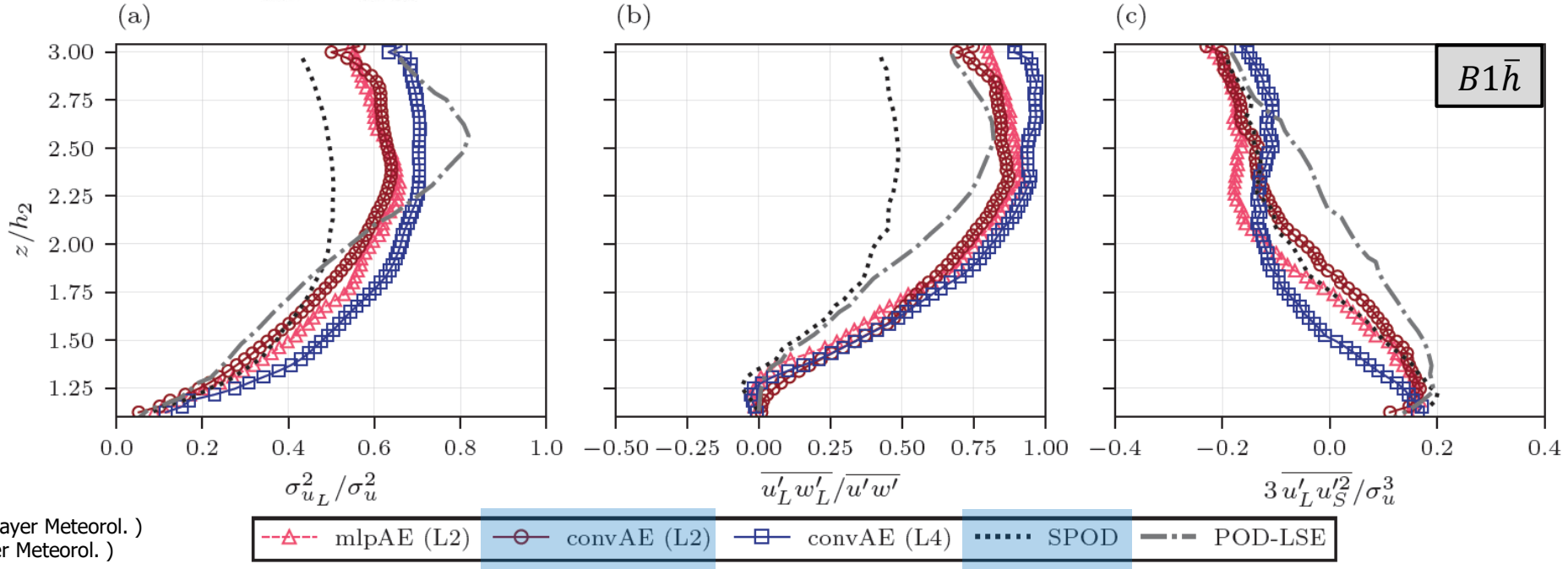
- Only look at YZ plane measurement for modelling procedure today
- Training/Validation/Test dataset split is around 70/15/15



> Methods > Field AE & SPOD selection



- Compared Multi-Layer Perceptron AE (mlpAE), convAE and SPOD .
- Retained Turbulent Kinetic Energy (TKE) is not linear with latent dimension (mlpAE and convAE).
- Hard to apply the same TKE threshold for convAE.
- Need to compare the statistics of interest.

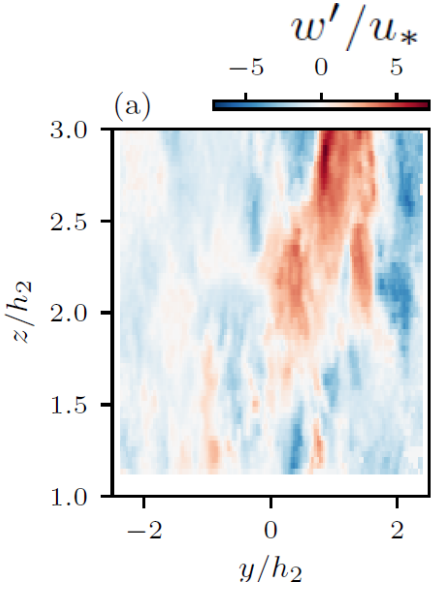


POD-LSE: (Du et al., 2024, Bound.-Layer Meteorol.)
SPOD: (Du et al., 2025, Bound.-Layer Meteorol.)

Results

> Results > Reconstruction of a snapshot

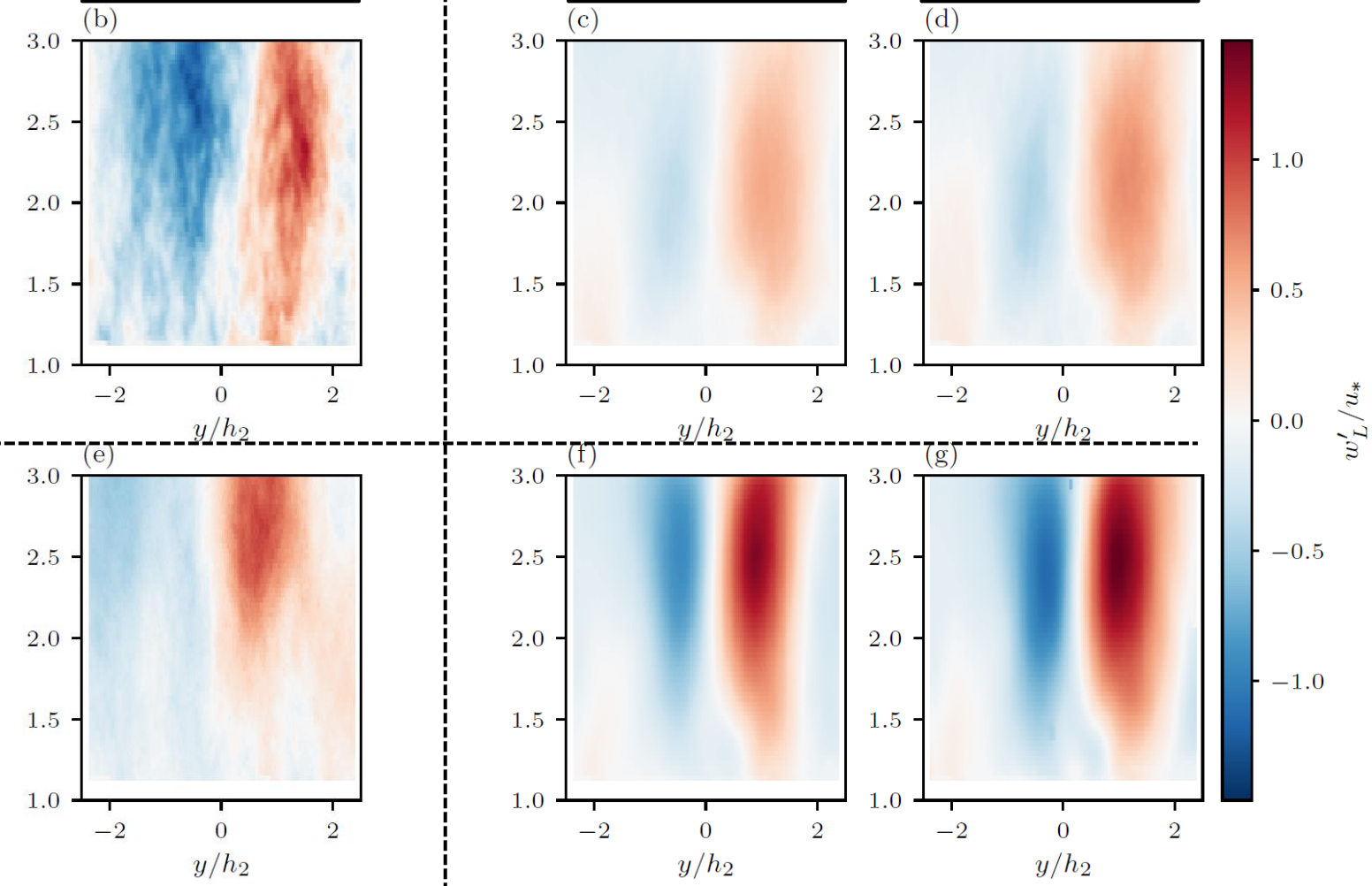
$B1\bar{h}$, test set, original full-scale



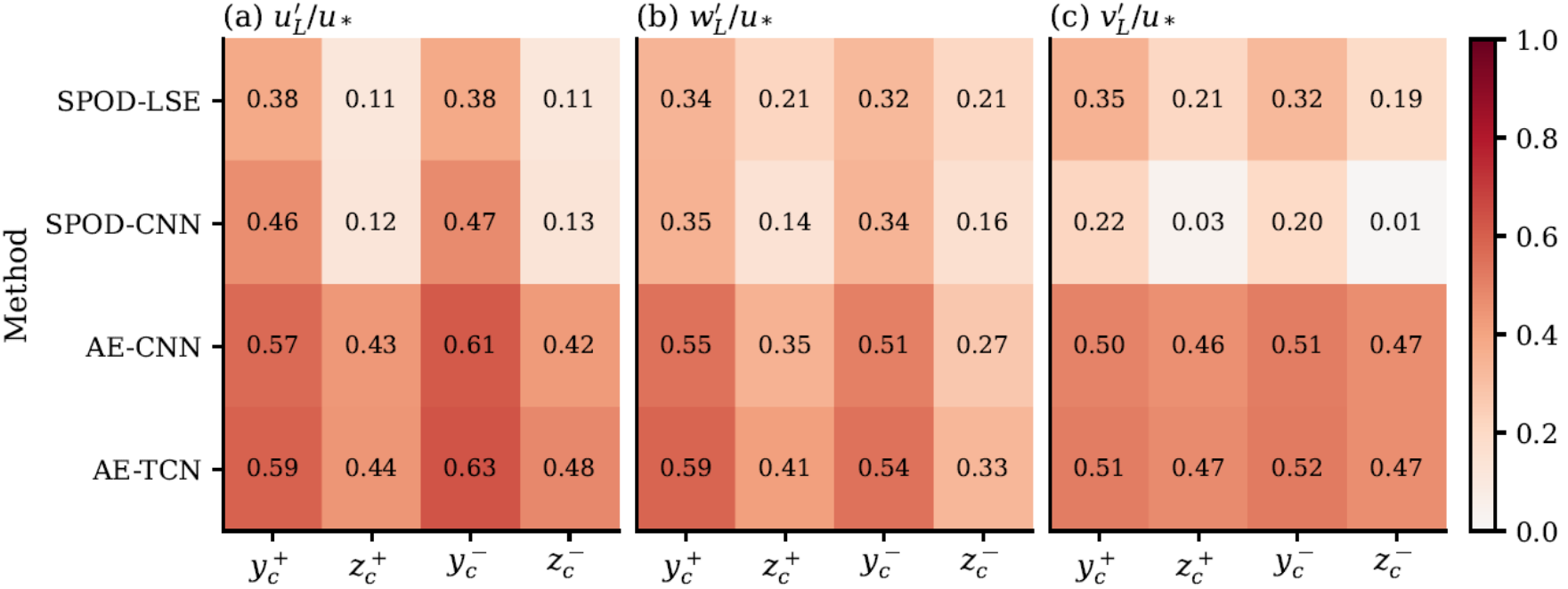
- **2** Scale separation techniques
  - SPOD
  - convAE
- **3** ROM
  - Linear Stochastic Esitmation (LSE)
  - Convolutional Neural Network (CNN)
  - Temporal CNN (TCN)

SPOD filtered | SPOD-LSE | SPOD-CNN

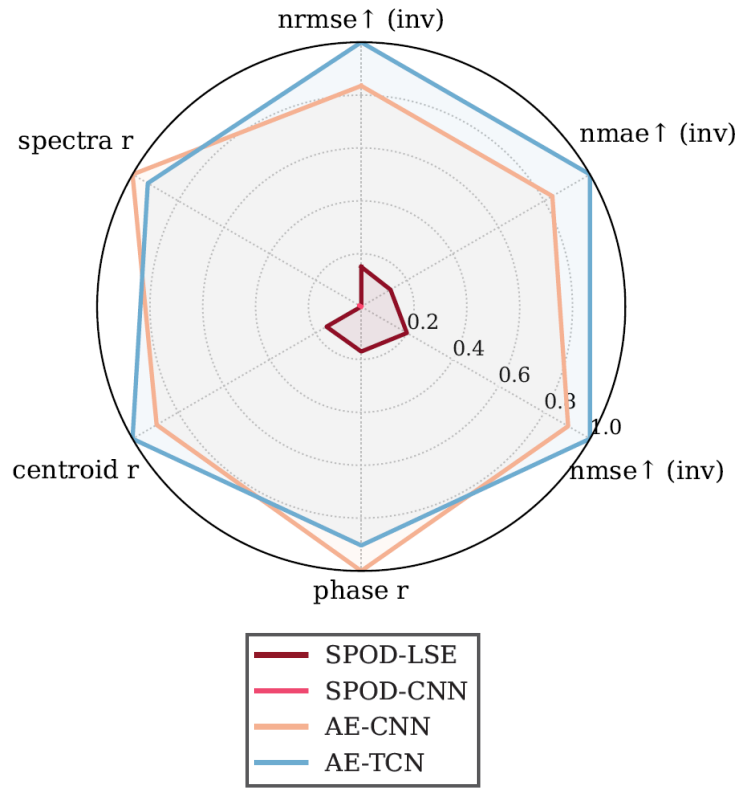
convAE filtered | convAE-CNN | convAE-TCN



$B1\bar{h}$ centroid correlations



Over all cases



- convAE-CNN tracks spectra and phase best
- convAE-TCN model gives the best balance

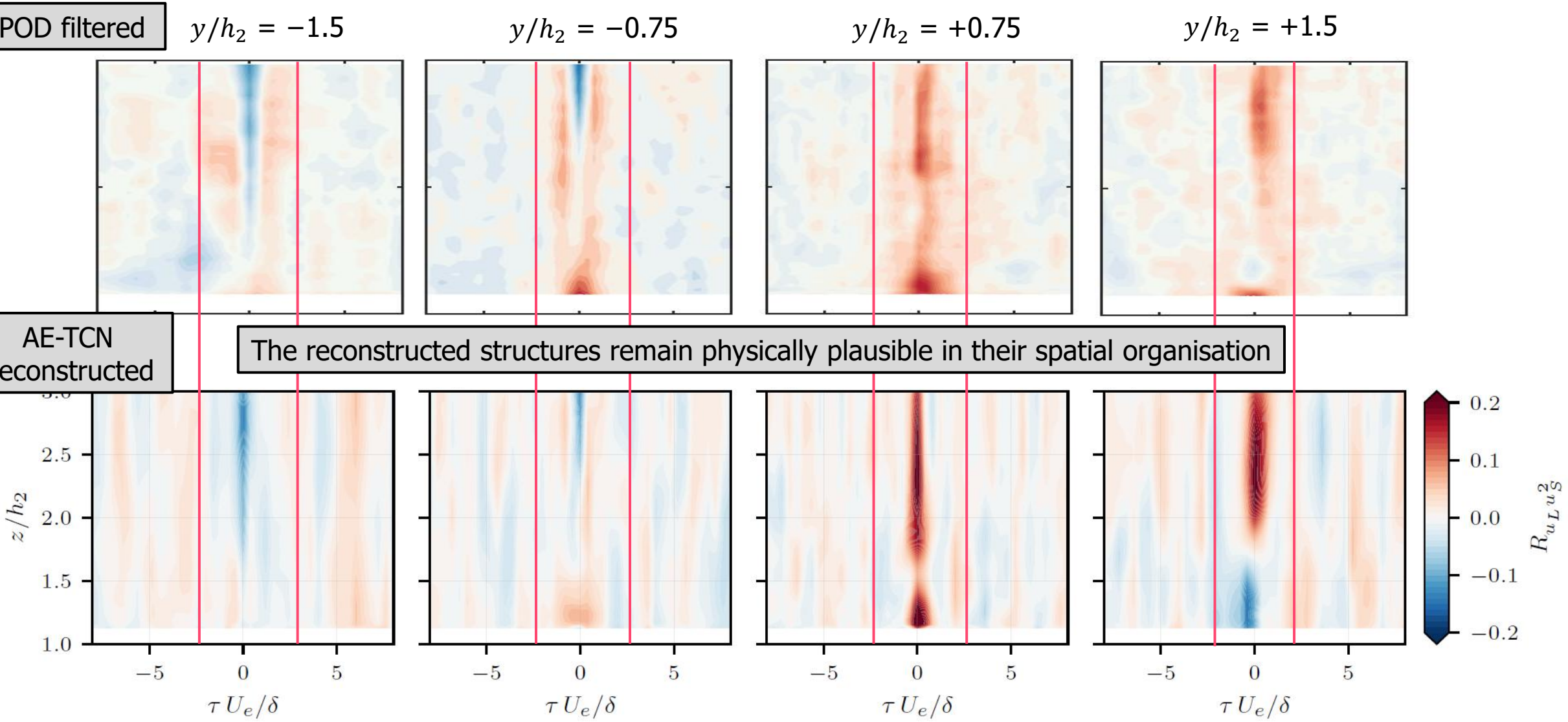
> Physical consistency check

Still need to check the physics of reconstructed large scales! Use $Cath$ case as example.

SPOD filtered

AE-TCN Reconstructed

The reconstructed structures remain physically plausible in their spatial organisation



Takeaways

> Takeaways

1. **Sparse synchronized measurements can bridge non-synchronized SPIV planes.**
   A common set of HWA measurements provides the link between independently acquired cross-plane datasets.
2. **Dominant large-scale flow organisation can be reconstructed from sparse measurements.**
   This provides a practical route to extract additional spatial information without simultaneous measurements.
3. **Nonlinear reconstruction improves the recovery of large-scale structures.**
   ConvAE-based models performed better overall than the linear baseline while retaining physically reasonable flow fields.
4. **Outlook:** QR-pivoting on a prior dataset could be used to select more informative probe locations in future experiments.
5. **Outlook:** 3D flow fields in Large Eddy Simulation may provide calibration for real 3D reconstruction.

NSERC CRSNG

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Dr. Eric Savory
Supervisor

anr agence nationale de la recherche

CENTRALE NANTES



Dr. Laurent Perret
International Advisor

Thank you

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PhD Candidate